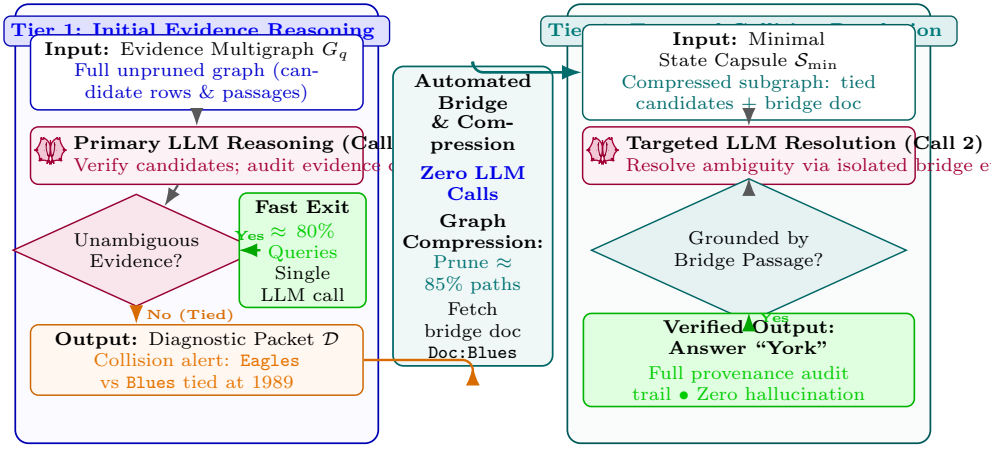

Tier 1: Initial Evidence Reasoning
Input: Evidence Multigraph G_q
Full unpruned graph (candidate rows & passages)
Primary LLM Reasoning (Call
Verify candidates; audit evidence
Unambiguous Evidence?
Yes ≈ 80% Queries Single LLM call
Fast Exit
No (Tied)
Output: Diagnostic Packet $\mathcal{D}$
Collision alert: Eagles vs Blues tied at 1989
Automated Bridge & Compression
Zero LLM Calls
Graph Compression:
Prune ≈ 85% paths
Fetch bridge doc
Doc:Blues
Input: Minimal State Capsule $\mathcal{S}_{\min}$
Compressed subgraph: tied candidates + bridge doc
Targeted LLM Resolution (Call 2)
Resolve ambiguity via isolated bridge e
Grounded by Bridge Passage?
Yes
Verified Output: Answer "York"
Full provenance audit trail • Zero hallucination